Spatial Estimates of SWE for the Rio Grande Basin

May 6, 2025

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Overview Map – Rio Grande Basin Study Area

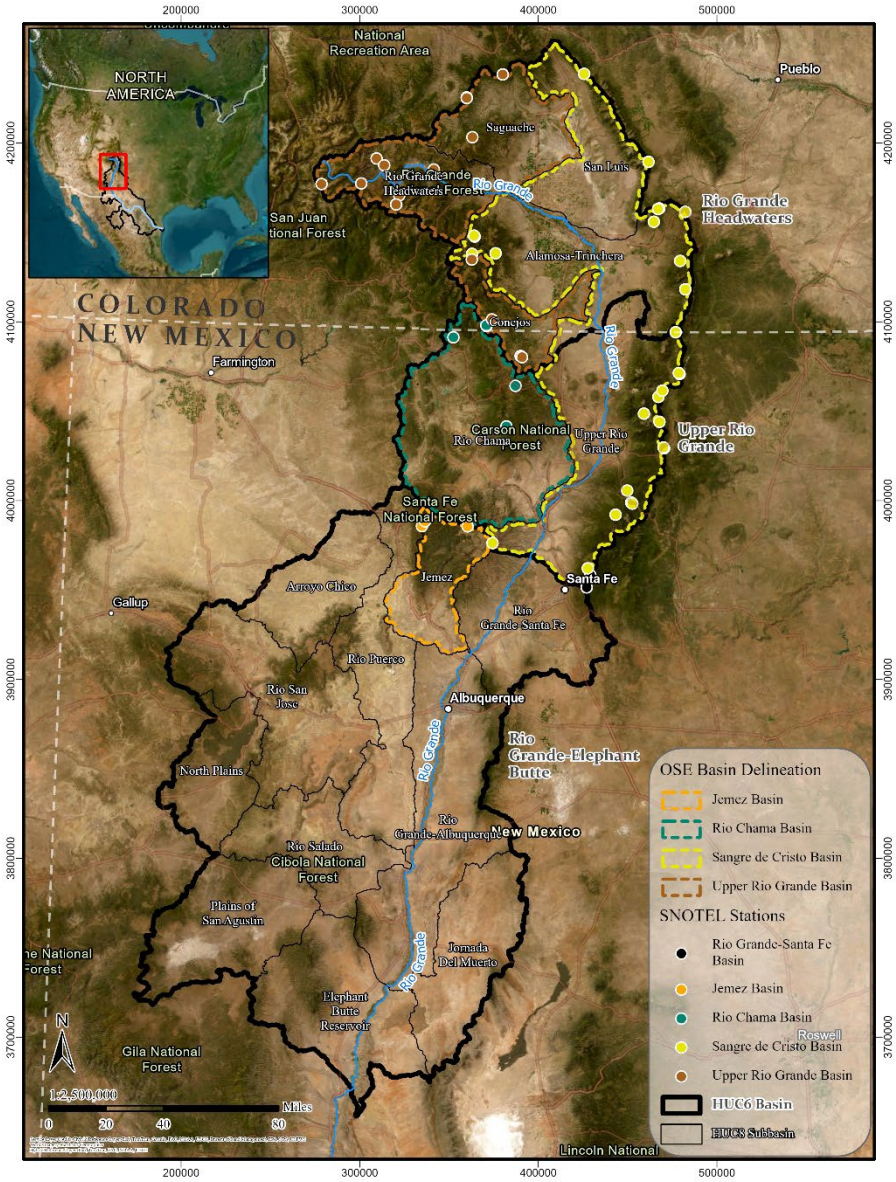

Figure 1. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – orange, Rio Chama – green, Sangre de Cristo – yellow, and Upper Rio Grande – brown) are symbolized.*

Summary of SNODAS SWE for May 6, 2025

- Another spring storm brought new snow to the region throughout the past few days (May 4 and 5, 2025) and likely into the next day or two.
- Spatial estimates of snow water equivalent (SWE) and change in SWE (presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 2.
- Due to the new snow, estimated SNODAS SWE increased in all three HUC6 Basins since the last report (April 29, 2025).
    - Estimated SNODAS SWE decreased in a single HUC8 subbasin across the entire RG Basin study area – the Rio Grande Headwaters (HUC8) subbasin (Tables 1 and 2; Figure 3).
    - The largest increase in estimated SNODAS SWE occurred across the Sangre de Cristo Mountains (Table 2).
    - The Jemez HUC8 subbasin had the largest increase in SNODAS SWE across the Rio Grande-Elephant Butte HUC6 Basin (Table 1; Figure 5).
- **Table 1**: Estimated SWE amount (median and mean in inches (in)), estimated volume (acre-feet (af)), and change in volume (af) since the last report, are shown for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Table 2**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown for the specific HUC8 subbasin groupings (OSE Basin Delineations shown in Figure 1; Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez).
- **Table 3**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown by basin and elevation band for the HUC6 basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Rio Grande Headwaters (HUC6)**: SNODAS SWE volume for May 6, 2025, is estimated to be 321,364 af, an increase of an estimated 74,207 af from April 29, 2025.
- **Upper Rio Grande (HUC6)**: SNODAS SWE volume for May 6, 2025, is estimated to be 243,305 af, an increase of an estimated 180,773 af from April 29, 2025.
- **Rio Grande-Elephant Butte (HUC6)**: SNODAS SWE volume for May 6, 2025, is estimated to be 13,386 af, an increase of an estimated 13,386 af from April 29, 2025.

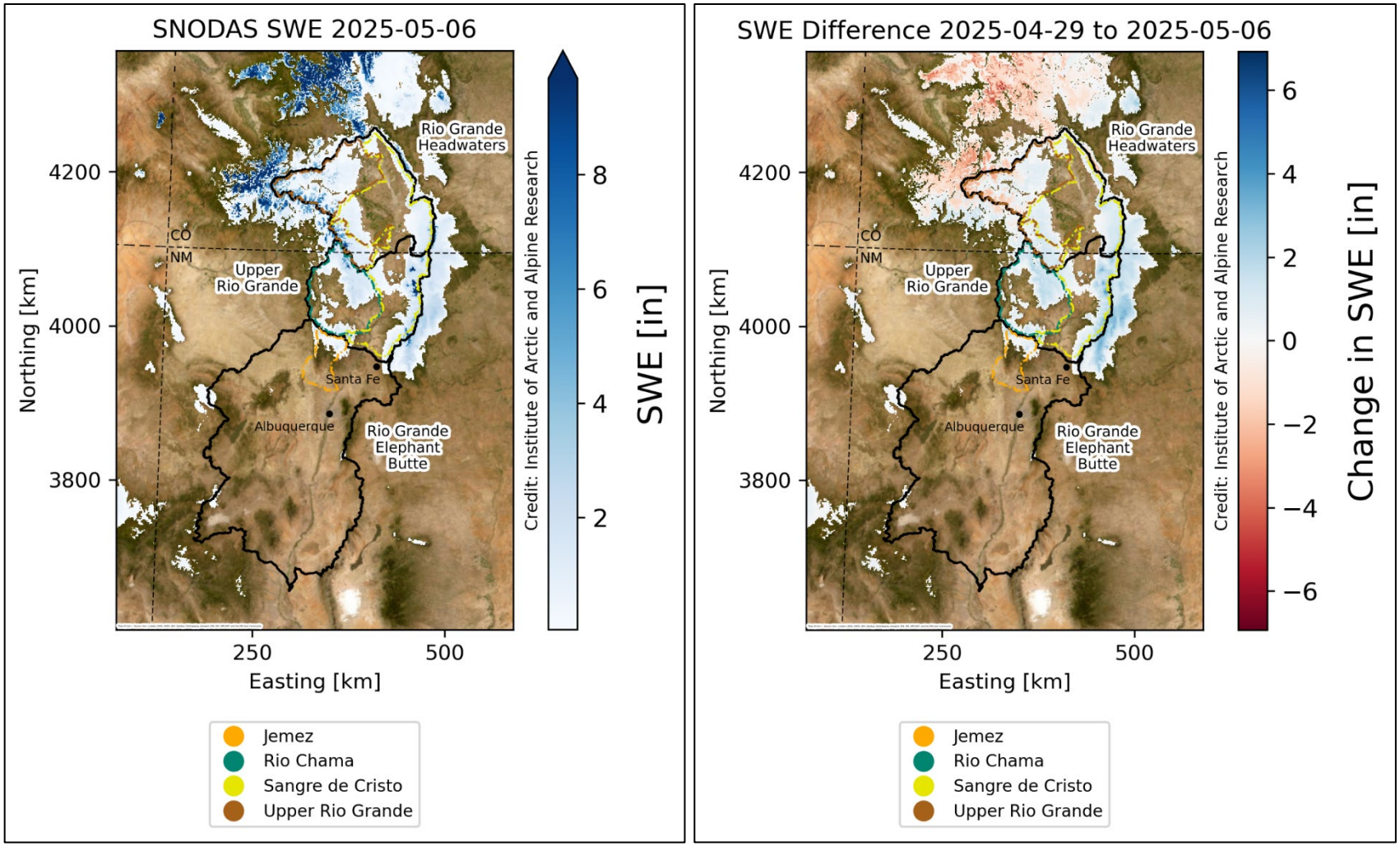

Figure 2. Estimated SNODAS SWE and Change in SNODAS SWE across the Rio Grande Basin study area. *SWE amounts (in) across the RG Basin study area for May 6, 2025 (left); dark blue pixels denote greater SWE amounts. Change in SNODAS SWE amounts (in) across the RG Basin study area from April 29, 2025, to May 6, 2025 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in) is presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

Figure 3 (Rio Grande Headwaters), Figure 4 (Upper Rio Grande), and Figure 5 (Rio Grande-Elephant Butte) show the spatial distribution of SWE amount (in), change in SWE volume (af) since the last report, and SWE volume (af) across subbasin and elevation. Since the last report, an uncharacteristically late spring storm brought new snow to the region. The Rio Grande Headwaters (HUC8) subbasin saw the only decrease in estimated SWE volume throughout the entire study region (Figure 3). Some other high elevation areas throughout the study region saw a decrease in estimated SWE volume, but overall almost all elevation bands across HUC8 subbasins saw an increase in estimated SWE. The Sangre de Cristo Mountains experienced the highest volume increase since the last report (Table 2), with parts of the Rio Grande-Elephant Butte (HUC6) Basin experiencing increased estimated SWE volumes as well (Table 1; Figure 5).

Estimated SWE amounts (in), volumes (af), and change in volumes (af) since the last report are presented in the tables below.

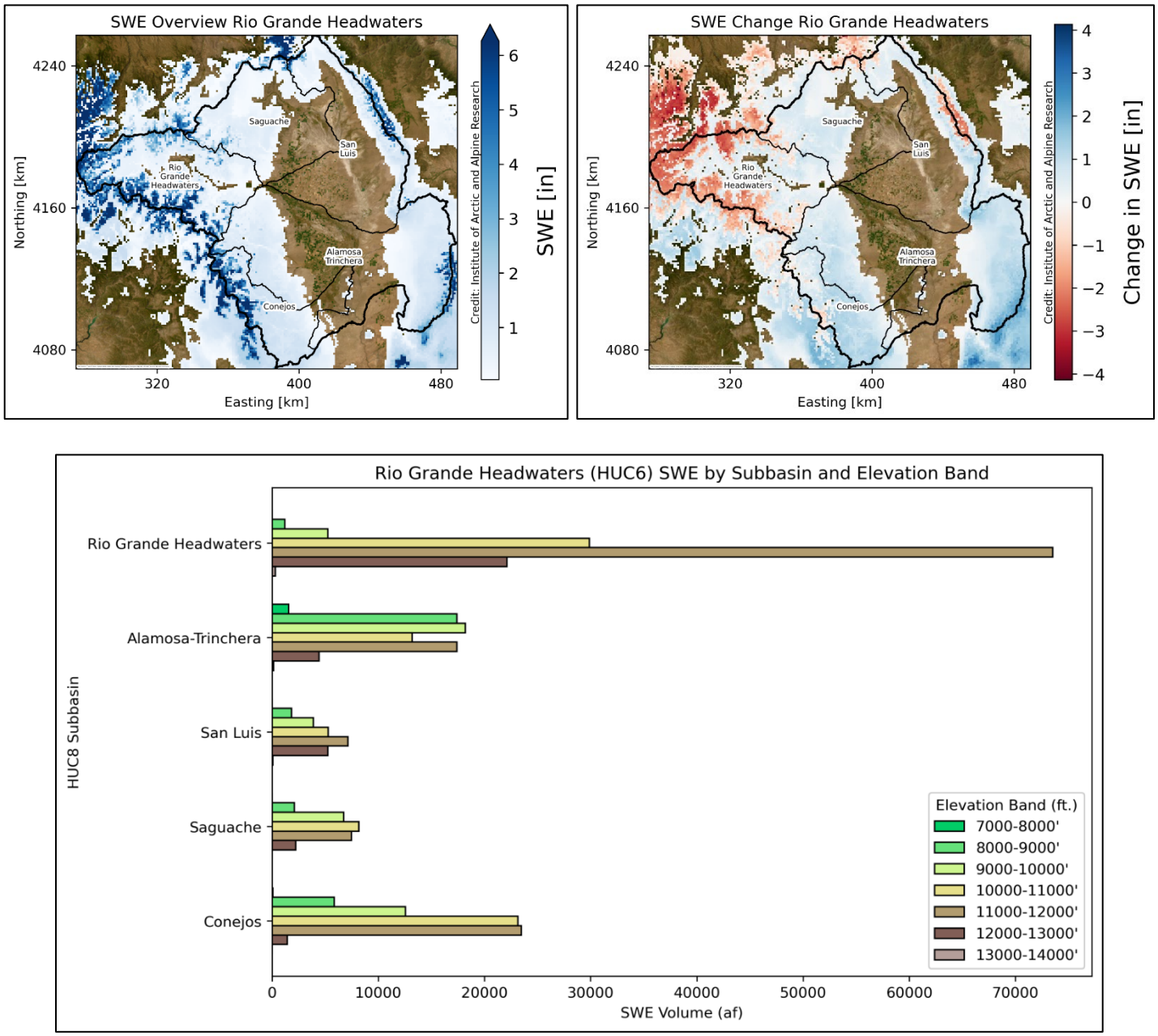

Figure 1. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande Headwaters Basin for May 6, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Rio Grande Headwaters Basin (HUC6), including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande Headwaters Basin (bottom).*

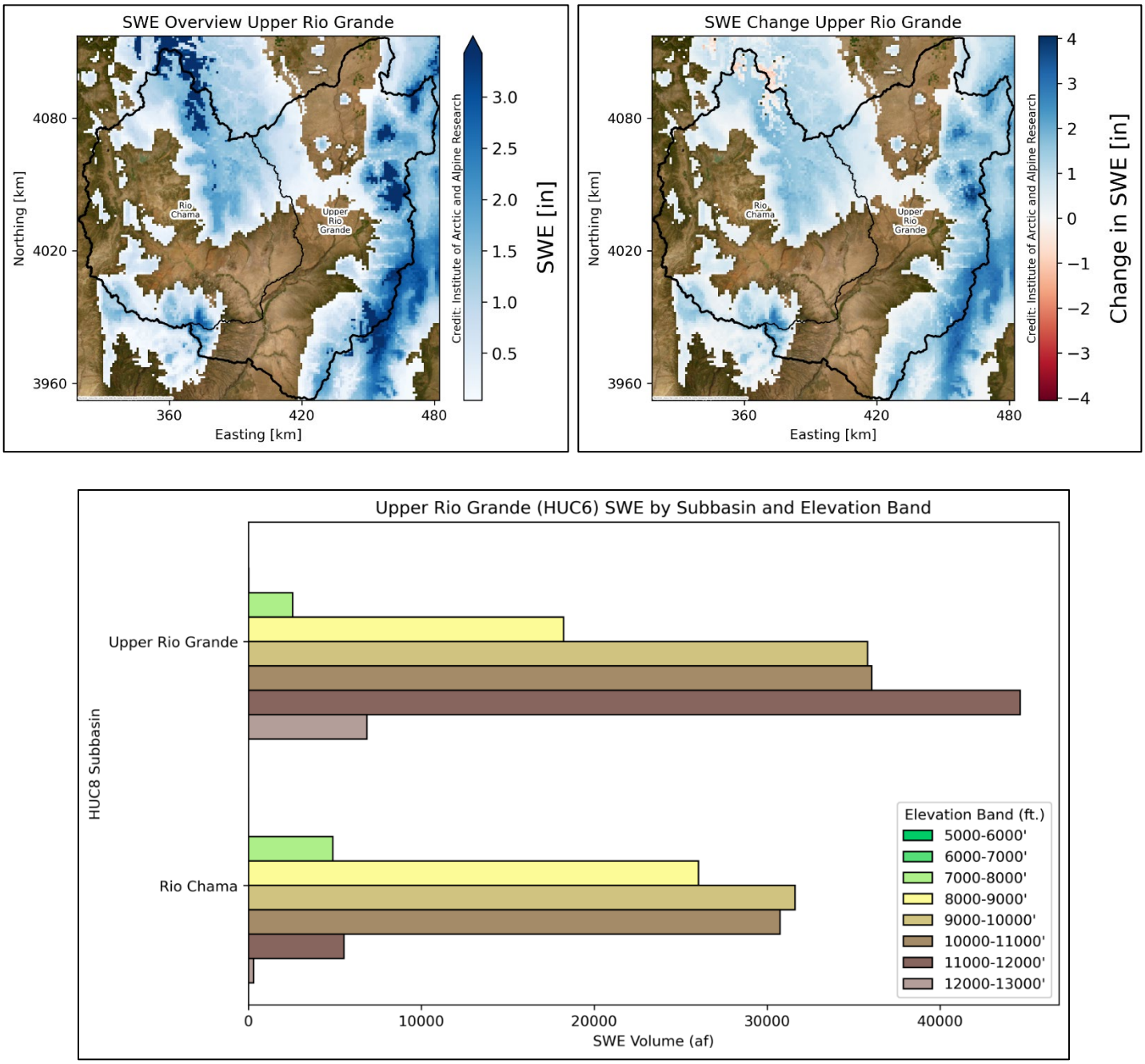

Figure 2. Estimated SNODAS SWE (in) and SWE volume (af) across the Upper Rio Grande Basin for May 6, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Upper Rio Grande Basin (HUC6), including the Upper Rio Grande and Rio Chama subbasins. SWE volume (af) by subbasin and elevation band across the Upper Rio Grande Basin (bottom).*

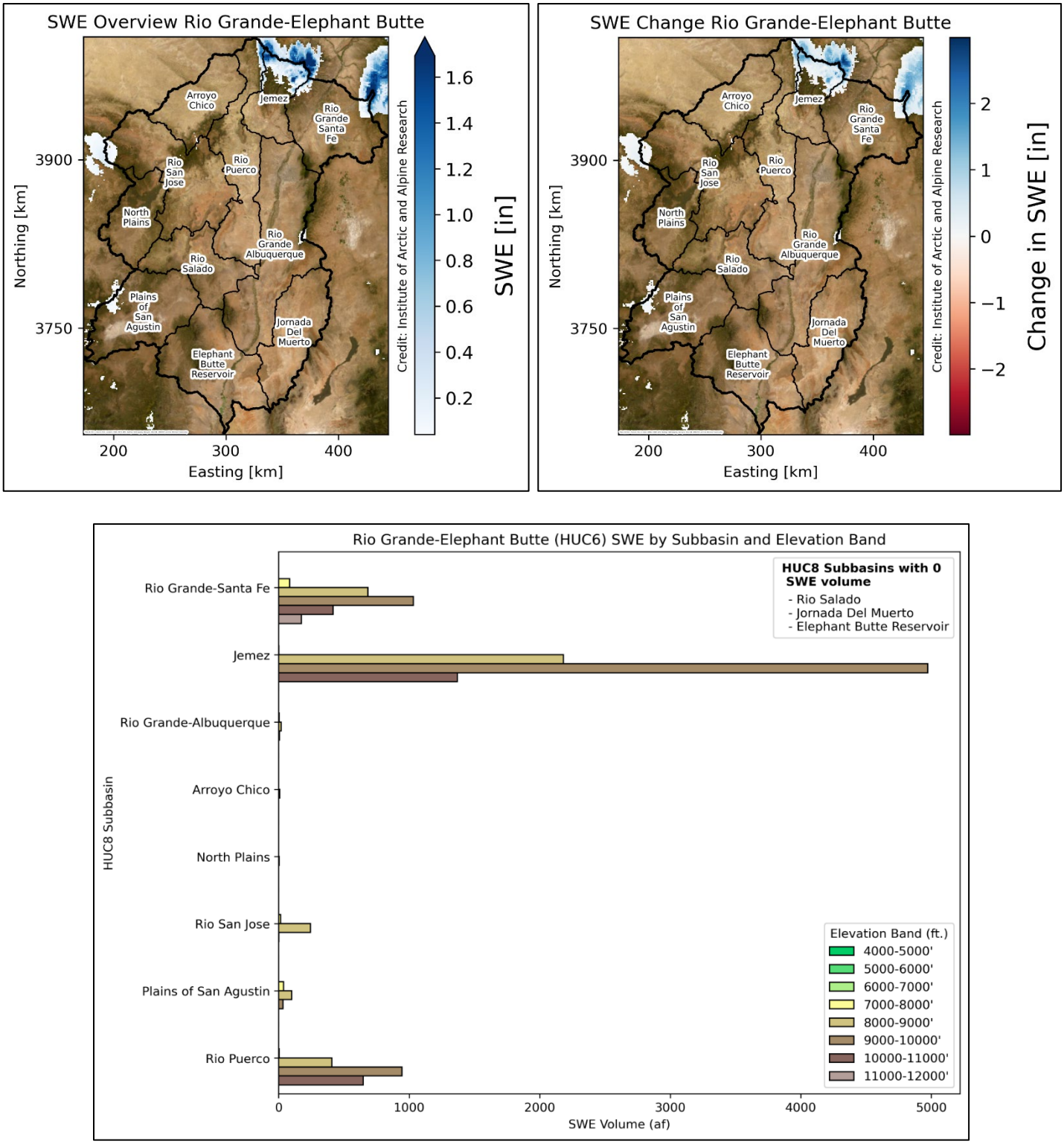

Figure 3. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande-Elephant Butte Basin for May 6, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across Rio Grande-Elephant Butte (HUC6), including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande-Elephant Butte Basin (bottom).*

Tables

Table 1. Estimated SWE by HUC6 Basin for May 6, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each HUC6 basin (bold).*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	0.6	1.8	132,309	-32,077
	Alamosa-Trinchera	0.3	0.5	72,303	59,819
	San Luis	0.0	0.3	23,459	8,512
	Saguache	0.2	0.4	26,752	12,650
	Conejos	1.1	1.6	66,542	25,303
	Total	**0.4**	**0.8**	**321,364**	**74,207**
Upper Rio Grande	Upper Rio Grande	0.2	0.8	144,174	106,761
	Rio Chama	0.1	0.6	99,131	74,012
	Total	**0.1**	**0.7**	**243,305**	**180,773**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	2,392	2,392
	Jemez	0.0	0.2	8,522	8,522
	Rio Grande-Albuquerque	0.0	0.0	29	29
	Arroyo Chico	0.0	0.0	7	7
	North Plains	0.0	0.0	4	4
	Rio San Jose	0.0	0.0	259	259
	Plains of San Agustin	0.0	0.0	170	170
	Rio Salado	0.0	0.0	0	0
	Jornada Del Muerto	0.0	0.0	0	0
	Elephant Butte Reservoir	0.0	0.0	0	0
	Rio Puerco	0.0	0.0	2,003	2,003
	Total	**0.0**	**0.0**	**13,386**	**13,386**

Table 2. Estimated SWE by Specified Subbasin Groupings for May 6, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
Upper Rio Grande	Rio Grande Headwaters	0.6	1.8	132,309	-32,077
	Saguache	0.2	0.4	26,752	12,650
	Conejos	1.1	1.6	66,542	25,303
	Total	**0.5**	**1.2**	**225,603**	**5,876**
Sangre de Cristo	San Luis	0.0	0.3	23,459	8,512
	Alamosa-Trinchera	0.3	0.5	72,303	59,819
	Upper Rio Grande	0.2	0.8	144,174	106,761
	Total	**0.1**	**0.6**	**239,936**	**175,092**
Rio Chama		**0.1**	**0.6**	**99,131**	**74,012**
Jemez		**0.0**	**0.2**	**8,522**	**8,522**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for May 6, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	11	11
		8000-9000'	0.1	0.1	1,205	1,205
		9000-10000'	0.4	0.4	5,253	4,504
		10000-11000'	0.6	1.4	29,895	-894
		11000-12000'	3.2	3.4	73,511	-26,461
		12000-13000'	3.6	3.7	22,128	-10,311
		13000-14000'	2.9	2.5	306	-130
	Alamosa-Trinchera	7000-8000'	0.0	0.0	1,578	1,578
		8000-9000'	0.5	0.5	17,389	17,389
		9000-10000'	0.9	0.9	18,190	18,169
		10000-11000'	1.0	1.1	13,189	11,441
		11000-12000'	1.5	2.0	17,391	8,357
		12000-13000'	2.2	2.4	4,419	2,742
		13000-14000'	1.8	1.8	148	144
	San Luis	7000-8000'	0.0	0.0	17	17
		8000-9000'	0.1	0.2	1,832	1,832
		9000-10000'	0.4	0.5	3,890	3,805
		10000-11000'	0.7	0.9	5,287	3,512
		11000-12000'	0.8	1.5	7,129	294
		12000-13000'	1.2	2.0	5,247	-980
		13000-14000'	0.9	0.9	58	32
	Saguache	7000-8000'	0.0	0.0	10	10
		8000-9000'	0.1	0.1	2,104	2,104
		9000-10000'	0.4	0.4	6,735	6,405
		10000-11000'	0.6	0.7	8,187	5,218
		11000-12000'	1.2	1.6	7,482	-1,008
		12000-13000'	2.2	2.4	2,216	-96
		13000-14000'	0.8	0.8	17	17
	Conejos	7000-8000'	0.0	0.0	72	72
		8000-9000'	0.7	0.7	5,843	5,837
		9000-10000'	1.1	1.2	12,560	11,441
		10000-11000'	1.4	2.4	23,137	5,816

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
		11000-12000'	5.1	4.8	23,491	1,941
	Conejos	12000-13000'	3.5	3.7	1,438	195
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	19	19
		7000-8000'	0.0	0.0	2,558	2,558
		8000-9000'	0.6	0.6	18,225	18,197
		9000-10000'	1.4	1.4	35,820	35,505
		10000-11000'	1.8	2.2	36,052	30,444
		11000-12000'	3.3	5.7	44,643	17,176
		12000-13000'	4.1	5.8	6,857	2,861
	Rio Chama	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.1	4,890	4,890
		8000-9000'	0.7	0.8	26,030	25,899
		9000-10000'	1.5	1.5	31,611	29,355
		10000-11000'	1.7	2.7	30,751	13,234
		11000-12000'	7.1	6.9	5,535	599
		12000-13000'	7.6	7.6	315	36
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	84	84
		8000-9000'	0.1	0.1	684	684
		9000-10000'	0.9	0.9	1,032	1,032
		10000-11000'	1.8	1.7	418	418
		11000-12000'	2.1	2.1	174	174
	Jemez	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	2	2
		8000-9000'	0.1	0.1	2,181	2,181
		9000-10000'	0.7	0.9	4,971	4,971
		10000-11000'	1.3	1.4	1,368	1,368
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	4	4

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
		8000-9000'	0.0	0.0	19	19
Rio Grande-Elephant Butte	Rio Grande-Albuquerque	9000-10000'	0.0	0.0	6	6
		10000-11000'	0.0	0.0	0	0
	Arroyo Chico	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	7	7
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	North Plains	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	4	4
	Rio San Jose	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	15	15
		8000-9000'	0.0	0.0	243	243
		9000-10000'	0.0	0.0	1	1
		10000-11000'	0.0	0.0	0	0
	Plains of San Agustin	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	36	36
		8000-9000'	0.0	0.0	101	101
		9000-10000'	0.0	0.1	32	32
	Rio Salado	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
	Jornada Del Muerto	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 29, 2025 (af)
		6000-7000'	0.0	0.0	0	0
Rio Grande-Elephant Butte	Elephant Butte Reservoir	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	Rio Puerco	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	4	4
		8000-9000'	0.0	0.1	407	407
		9000-10000'	0.9	0.9	944	944
		10000-11000'	1.2	1.2	649	649

Acknowledgments

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